

Huawei

AI Infrastructure Architecture Evolution

2015-2026

Kunpeng · Ascend · UnifiedBus · DPU · AI NIC
SuperPoD · Optics · Storage · Software

A generation-by-generation synthesis of conference disclosures and primary technical sources.
Evidence cutoff: 28 September 2026.
Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and executive synthesis

Coverage: Kunpeng host CPUs and Ascend NPUs receive deep coverage; UnifiedBus, DPUs, AI NICs, systems and software receive full generation-level coverage.

Period: 2015-28 September 2026; latest verified event: HUAWEI CONNECT 2026.

Focus: architecture, implementation and the balance between compute, memory and communication.

Related lines: Atlas edge inference, HiSilicon networking, optical engines and KV-cache storage where they explain the server stack.

Exclusions: an exhaustive Kirin smartphone, consumer GPU, telecom baseband or storage-drive catalogue; unverified manufacturing and capacity estimates.

Huawei's infrastructure evolved through three architectural commitments. First, it built custom Arm server cores and reusable compute/I/O dies, culminating in the 2019 Kunpeng 920. Second, Ascend separated matrix, vector and scalar work and made explicit data movement central to performance. Third, the system boundary expanded from a server to a switched pool of CPUs, NPUs and memory. CloudMatrix384 made that third commitment concrete; Ascend 950 and UnifiedBus 2.0 extend it with dedicated communication engines and more flexible memory semantics.

The strongest current technical disclosure is the Ascend 950 architecture white paper. It describes two compute dies, two I/O dies, up to 36 matrix cores and 72 vector cores, a coherent on-package memory organization and up to 128 MB of accelerator L2. The PR and DT branches explicitly trade memory cost and bandwidth against workload needs. PR targets prefill and recommendation; DT targets training and complex inference. Atlas 350 is a concrete 950PR card configuration, not the maximum specification of the entire chip family.

The September 2026 roadmap brings Ascend 960DT readiness forward to Q1 2027 and 960PR to Q3 2027. Atlas 960E is an announced 4,096-NPU, liquid-cooled NPO system; its launch is not evidence that 960 silicon is already generally available. The engineering direction is clear: optical reach, collective offload and a hierarchy of HBM, CPU memory and KV storage increasingly determine usable model throughput. At the same time, the public record remains thinner on recent Kunpeng core internals, foundry processes and measured system power than on Ascend programming and system topology.

Flagship milestones and comparable quantities

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Milestone	CPU / memory	Accelerator matrix BF16 / FP8	Accelerator memory	Scale-up scope
2019	Kunpeng 920: 64 cores; 187.7 GB/s DDR4	Ascend 910: BF16 n/d; FP16 256 TFLOPS	4 HBM2 stacks; ~1.2 TB/s	HCCS-connected accelerator nodes
2025	CloudMatrix384: 192 Kunpeng CPUs	910C reference: BF16 supported; peak n/d in reviewed system paper	2 compute dies, 8 memory stacks per NPU	384 NPUs
950PR	8C/16T Linx816 on maximum NPU configuration	432 / 865 TFLOPS	128 GB; 1.6 TB/s maximum	8,192 architectural domain; products vary
950DT	Host Kunpeng 950: 64/96-core server configurations	486 / 973 TFLOPS	144 GB; 4 TB/s	8,192 architectural domain; products vary
960 / 960E	Kunpeng memory pools accompany NPU domains	8 EFLOPS FP8 per announced 960E system	Up to 1 PB per system	4,096 NPUs

02 Reading the numbers and product names

Bandwidth uses decimal GB/s and TB/s. Ethernet Gb/s is divided by eight for a raw one-direction byte rate; packet, encoding and protocol overheads remain. A full-duplex total is halved only when the source explicitly identifies it as bidirectional. Peak compute counts a multiply-add as two operations. Matrix-only throughput is kept separate from the sum of Cube and Vector throughput. No sparse multiplier is applied. Precision labels remain exact: FP16 is not BF16, INT8 is not FP8, and MXFP4 is not a generic interchangeable four-bit format.

Ascend names a processor family, DaVinci an architecture, Atlas hardware products, and CloudMatrix a system/cloud architecture. Kunpeng is the CPU family; TaiShan is also used for server products, while TaiShan V110 denotes the 920 core. Unified Buffer and UnifiedBus both abbreviate to UB: the former is local AI-core storage and the latter is the system interconnect. A product marked n/d has a material specification absent from the reviewed evidence; this is not a zero or proof that the feature does not exist.

The bibliography is connected to chapters through a compact source map. Hot Chips 2019 supplies the original DaVinci organization; the HPCA 2021 industry paper and Huawei's published technical collection add programming and implementation context. IEEE Micro's Kunpeng paper supplies CPU internals. IEEE Micro is a journal, not the MICRO conference. Related ASPLOS, MICRO and ISCA research is included without treating experimental proposals as shipping circuitry. Announced capabilities, architectural maxima and measured deployments retain their separate scopes.

03 Host CPU generations

Kunpeng: reference configurations and disclosure boundaries · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Codename	GA	Status	Core march	Cores / threads
Hi1610 / Hi1612 baseline	Hi1610 / Hi1612	2015 driver record; GA n/d	Historical shipping	n/d	n/d
Kunpeng 916	Hi1616	Pre-2019 generation	Historical shipping	n/d	32 / 32
Kunpeng 920	Hi1620	2019	Historical shipping	TaiShan V110	64 / 64
Kunpeng 920 new models	n/d	Current motherboard portfolio; GA n/d	Shipping / available	n/d	SMT2; SKU core count n/d here
Kunpeng 950	n/d	2026 H1 server brief	Shipping / available	n/d	64 / 96 cores per socket; SMT2
Kunpeng 950 larger configuration	n/d	2025 roadmap; GA n/d	Roadmap	n/d	192 / 384

Kunpeng: reference configurations and disclosure boundaries · 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
Hi1610 / Hi1612 baseline	n/d	n/d	n/d	n/d
Kunpeng 916	16 nm	n/d	n/d	n/d
Kunpeng 920	Compute 7 nm; I/O 16 nm	2 compute + 1 I/O; CoWoS reference	512 KiB	64 MB; sliced, ~1 MB/core
Kunpeng 920 new models	n/d	n/d	n/d	n/d
Kunpeng 950	n/d	n/d	n/d	n/d
Kunpeng 950 larger configuration	n/d	n/d	n/d	n/d

Kunpeng: reference configurations and disclosure boundaries · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
Hi1610 / Hi1612 baseline	n/d	n/d	n/d	n/d	n/d
Kunpeng 916	DDR4; 4 channels; 2400 MT/s; 76.8 GB/s	2P boards	PCIe 3.0; CXL n/a	n/d	Arm NEON
Kunpeng 920	DDR4; 8 channels; 2933 MT/s; 187.7 GB/s	2P / 4P; HCCS	PCIe 4.0 ×40; CXL n/a	200 W paper reference	128-bit NEON
Kunpeng 920 new models	DDR5; channels / bandwidth n/d	2P / 4P boards	PCIe 5.0; CXL n/d	n/d	n/d
Kunpeng 950	DDR5 RDIMM 7200 / MRDIMM 8800; channels n/d	2P reference board	n/d	n/d	n/d
Kunpeng 950 larger configuration	n/d	n/d	n/d	n/d	n/d

04 2015-2019: from server Arm to Kunpeng 920

The 2015 Linux HNS driver submission establishes Hi1610 and Hi1612 as an early server-platform baseline. Kunpeng 916 then used 32 Arm cores and a DDR4 platform. That generation's significance was platform integration: boot firmware, network drivers, memory validation and operating-system enablement made the later custom-core server product practical. It is useful context, but the reviewed platform documents do not establish the same core implementation as the custom V110 introduced with Kunpeng 920.

Kunpeng 920's TaiShan V110 is an out-of-order Arm core with a four-instruction front end/dispatch organization and a multilevel branch-prediction structure. The published design provides separate 64 KiB instruction and data L1 caches and 512 KiB of private L2. Its floating-point/vector units implement 128-bit NEON operations; it should not be assigned Neoverse or SVE characteristics merely because it executes the Arm ISA. Public detail does not establish a complete reorder-buffer, scheduler, load/store queue or branch-predictor capacity inventory.

The LLC averages about 1 MB per core in the 64-core configuration. Tags and distributed data slices connect through the on-die fabric; the paper organizes coherence around home agents rather than a single large centralized cache. The dual-ring fabric emphasizes predictable service and throughput. These details matter because a server's sustainable performance depends on where a cache miss is resolved and how memory traffic crosses compute-die boundaries, not simply on the sum of cache capacity.

The high-end reference combines two 32-core 7 nm compute dies with a 16 nm compute-I/O die. The paper describes CoWoS assembly and a roughly 400 GB/s coherent die-to-die connection; its bandwidth direction is not specified and is therefore not halved in this report. A single-compute-die Lite configuration demonstrates the reuse strategy. Compute and I/O technology can be selected separately, and the same I/O building blocks feed other HiSilicon products. The cost is that package links, memory placement and NUMA policy become first-class design choices.

Eight 64-bit DDR4-2933 channels yield a theoretical 187.7 GB/s, or 2.93 GB/s per core at 64 cores. The CPU also offers PCIe 4.0 and integrated high-speed Ethernet options. UMA and NUMA presentation modes are both discussed in the technical record; one address space should not be read as identical physical latency to every memory location. The practical AI-host role is data preparation, orchestration, networking and memory staging, while dense matrix work belongs on Ascend.

05 Updated 920 and Kunpeng 950: platform evolution

The current motherboard portfolio explicitly distinguishes "new-model Kunpeng 920" from the original DDR4 product. New-model boards support simultaneous multithreading, DDR5 and PCIe 5.0, including two- and four-socket configurations. The 920 brand therefore spans materially different platforms. It would be incorrect to combine the original V110 core/cache disclosure with the newer board's memory and I/O and call that a fully documented single SKU.

The 2026 H1 Kunpeng 950 board brief is more concrete than the earlier roadmap: it specifies two processors with 64- or 96-core options, a 2.3 GHz base clock with Turbo, up to 24 DIMMs, DDR5 RDIMM operation up to 7200 MT/s and MRDIMM operation up to 8800 MT/s. Its 3 TB capacity is a two-socket system configuration. Twenty-four DIMM sockets do not establish twelve memory channels per processor; the document does not disclose the necessary channel-to-slot mapping.

The platform emphasizes Arm CCA, secure device access, trusted measurement and RAS, including online handling of faulty cores. These are important when a CPU hosts multi-tenant AI services or provides memory to other processors. They do not establish a confidential-computing guarantee for every external NPU, NIC and software path. The earlier 192-core/384-thread 950 configuration remains separately identified as a roadmap item unless a matching shipping SKU and system are documented.

Kunpeng's emerging role extends beyond feeding an accelerator over PCIe. CloudMatrix and the later Peerium architecture expose CPU-attached memory as a resource for the wider compute domain. This can improve capacity efficiency for embedding tables, KV state and agent sandboxes, but locality still matters: remote bandwidth, synchronization, ownership and failure recovery have to be handled by the runtime. A large pool is a capacity resource before it is a substitute for local HBM bandwidth.

06 Ascend generation comparison

Ascend NPUs: chip and card scopes are explicit · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging	Compute units
Ascend 310	DaVinci	2018	Historical shipping	12 nm	n/d	2 AI cores
Ascend 910	DaVinci	2019	Historical shipping	Compute N7+; other die nodes n/d	Compute + Nimbus I/O + 4 HBM2 + dummy dies	32 AI cores
Ascend 310P	Inference branch	Atlas 300I Pro / Duo generation	Historical shipping	n/d	n/d	n/d
Ascend 910B2 / A2	DaVinci	2023-era A2 systems	Historical shipping	n/d	n/d	24 AIC + 48 AIV
Ascend 910C / A3	DaVinci	2025-03	Shipping / available	n/d	2 compute dies; 8 memory stacks	48 AIC + 96 AIV
Ascend 950PR	DaVinci 3	2026; Atlas 350 product	Shipping / available	n/d	2 AI + 2 I/O dies; 8 memory modules	32 AIC + 64 AIV
Atlas 350	950PR / DaVinci 3	2026 product listing	Shipping / available	n/d	950PR card configuration	28 AIC + 56 AIV
Ascend 950DT	DaVinci 3	2026 systems listed; earlier Q4 target	Product-listed; chip GA date n/d	n/d	2 AI + 2 I/O dies; 4 memory modules	36 AIC + 72 AIV
Ascend 960DT / PR	n/d	DT Q1 2027; PR Q3 2027 readiness	Roadmap	n/d	n/d	n/d

Ascend NPUs: chip and card scopes are explicit - 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix	Memory: type; stacks; GB; TB/s	LLC / SRAM
Ascend 310	n/d	n/d	n/d	n/d	External LPDDR; board-dependent	n/d
Ascend 910	n/d; FP16 256 TFLOPS	n/d	n/d	n/d	HBM2; 4 stacks; capacity SKU-dependent; ~1.2 TB/s	L2 fabric 4 TB/s in HC talk
Ascend 310P	n/d	n/d	n/d	n/d	LPDDR; configuration-dependent	n/d
Ascend 910B2 / A2	Supported; SKU peak n/d	n/d	n/d	n/d	64 GB HBM in ENEC 910B2 test system; rate n/d	n/d
Ascend 910C / A3	Supported; peak n/d in R02	INT8 path; native FP8 not established	n/d	n/d	8 stacks; capacity / rate n/d in R02	n/d
Ascend 950PR	432 TFLOPS	865 TFLOPS	MXFP4 1730 TFLOPS; FP6 n/d	n/d	128 GB; 1.6 TB/s; stack generation n/d	128 MB
Atlas 350	378 TFLOPS	756 TFLOPS	MXFP4 1513 TFLOPS	n/d	112 GB HBM; 1.4 TB/s	112 MB
Ascend 950DT	486 TFLOPS	973 TFLOPS	MXFP4 1946 TFLOPS	n/d	144 GB max; 4 TB/s; 96 GB variant	128 MB
Ascend 960DT / PR	n/d	n/d	n/d	n/d	n/d	n/d

Ascend NPUs: chip and card scopes are explicit - 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Scale-up: links; GB/s per direction; domain	Host link	Power / cooling	Form factor
Ascend 310	n/d	PCIe	8 W chip	Chip / Atlas modules
Ascend 910	3 × 240 Gb/s HCCS; direction n/d	PCIe 4.0	350 W HC target; launch 310 W	Atlas accelerator / server
Ascend 310P	n/d	PCIe	n/d	Inference cards / edge systems
Ascend 910B2 / A2	HCCS; configuration-dependent	PCIe	n/d	Atlas A2 servers / modules
Ascend 910C / A3	7 UB transceivers per die; 384-NPU domain	Kunpeng / UB system	n/d	Atlas 900 A3 / CloudMatrix384
Ascend 950PR	18 ×4 ports; 56 GB/s raw/link/direction; 1008 GB/s total/direction before port reuse	PCIe 5.0 ×16	n/d	Chip; several bins
Atlas 350	4 cards: 159 GB/s/direction/card; 2 cards: 212 GB/s/direction/card	PCIe 5.0 ×16	≤600 W; passive heatsink	PCIe OH FL DW
Ascend 950DT	18 ×4 ports; 56 GB/s raw/link/direction; 1008 GB/s total/direction before port reuse	PCIe 5.0 ×16	n/d	Chip; several bins
Ascend 960DT / PR	n/d	n/d	n/d	Future chips; Atlas 960E system announced

The 950 table uses Cube-only throughput, at the largest disclosed bin for each chip and the actual 28-Cube bin for Atlas 350. The Atlas 350 product page instead advertises 425 TFLOPS BF16, 804 TFLOPS MXFP8/HiF8 and 1,561 TFLOPS MXFP4 by combining Cube and Vector capability. The white paper gives 378, 756 and 1,513 TFLOPS respectively for its Cube engines. These are different accounting bases, not a contradiction. A quoted mixed-engine peak is not necessarily simultaneously attainable by one GEMM kernel.

07 2018-2021: DaVinci and the first Ascend systems

DaVinci's original design separates scalar control, vector operations and a Cube matrix engine. A $16 \times 16 \times 16$ FP16 multiplication produces 4,096 multiply-accumulates, conventionally counted as 8,192 floating-point operations per cycle. That headline rate requires operands to be tiled and reused through local buffers. MTE data-movement engines, L1 storage, L0 operand/result buffers and the Unified Buffer form an explicitly managed hierarchy. Overlapping transfer, matrix work and vector epilogues is the central programming task.

Ascend 310 put that architecture into an approximately 8 W inference chip with two AI cores, up to 16 TOPS INT8 or 8 TFLOPS FP16. Atlas modules and cards coupled inference with video preprocessing and host integration. The point of covering 310 alongside a datacenter product is architectural continuity: the local-buffer and operator-programming model scales across power envelopes. Mobile SoCs using DaVinci blocks and later 310P edge products extend this lineage, but their CPU and memory subsystems are different products.

Ascend 910 scaled to 32 AI cores and a package combining a 456 mm^2 compute die, an I/O die and four HBM2 stacks. The Hot Chips talk shows a 6×4 on-chip mesh, roughly 1.2 TB/s external memory bandwidth and HCCS plus Ethernet interfaces. Its compute-die area must not be confused with the sum of all die areas in the package. Likewise, the same talk's concept combining logic, 3D SRAM and twelve HBM stacks is a forward-looking packaging illustration, not the shipping 910 configuration.

Hot Chips used a 350 W design target, while the August 2019 launch reported a maximum of 310 W for 256 TFLOPS FP16 and 512 TOPS INT8. The difference is a disclosure update, not permission to apply 310 W to every later 910 board. The HPCA 2021 industry account explains how the hardware and software stack serve multiple AI scenarios. Taken together, these disclosures establish a complete early architecture story: specialized execution, explicitly controlled storage, heterogeneous packaging and a compiler/runtime that must schedule the resulting parallel engines.

08 A2 to A3: accelerator nodes become a compute pool

Atlas A2 products and the 910B family represent the intermediate server generation. Processor suffixes, enabled cores, memory configurations and server names vary; a single enthusiast specification table is not a safe substitute for an exact product document. The ISCA 2026 ENEC paper identifies its 910B2 test device as 24 Cube and 48 Vector units with 64 GB HBM, a useful fixed reference within that broader family. The ASPLOS 2025 study is valuable because it measures Ascend operator behavior and develops a component-level roofline model. It shows why peak matrix throughput alone misses serial execution phases, pipeline imbalance and interference between computation and data movement. That is product characterization and optimization research, not a disclosure of an unrelated next-generation chip.

Huawei dates Atlas 900 A3 delivery to March 2025 and identifies its accelerators as Ascend 910C. The CloudMatrix384 paper describes the corresponding dual-compute-die NPU package with eight memory stacks. Each die has 24 matrix cores and 48 vector cores, seven scale-up transceivers and a separate scale-out interface. The package-level count is therefore 48 matrix and 96 vector cores. This is an important hierarchy change: matrix engines, compute dies, NPU packages and server nodes are four different counting units.

CloudMatrix384 contains 48 compute nodes, each with eight NPUs, four Kunpeng CPUs and seven on-board UB switches: 384 NPUs and 192 CPUs in total. Twelve compute racks and four communication racks house a two-tier fabric. Seven independent switching subplanes connect the nodes without L2 oversubscription in the published topology. The CPU and NPU resources share the UB scale-up plane; NPU scale-out uses a separate RoCE plane; a QingTian DPU provides the VPC and storage-facing plane. The published system does not collapse all three into one physical network.

CloudMatrix-Infer separates prefill, decode and cache resources and uses large expert-parallel domains for mixture-of-experts serving. The paper reports DeepSeek-R1 results including 6,688 prefill tokens/s/NPU for a 4K prompt and 1,943 decode tokens/s/NPU with a 4K cache and time per output token below 50 ms. A tighter 15 ms constraint gives 538 decode tokens/s/NPU. These are vendor-reported end-to-end measurements with specific quantization and scheduling choices. They demonstrate the importance of the throughput/latency operating point; they are not interchangeable with a chip's FLOPS or an uncontrolled GPU comparison.

09 Ascend 950: computation, memory and communication co-design

The 950 package separates two AI dies from two I/O dies. PR uses eight memory modules; DT uses four higher-bandwidth modules. The maximum physical organization has 36 AI subsystems, each containing one Cube and two Vector cores, while commercial bins disable different amounts of compute and memory. Eight Linx816 Armv8-A CPU cores with two threads each supply local control and CPU operators. Their own caches are distinct from accelerator L2; hardware coherence connects the CPU and AI-core memory views inside the package.

DaVinci 3 strengthens work around the matrix multiply. Register-based vector execution, dual issue and SIMD/SIMT function blocks address elementwise arithmetic, gather/scatter and branch-heavy code. SIMD remains the principal throughput mode; SIMT improves irregular kernels rather than turning the entire NPU into a conventional GPU. A direct Cube-to-Vector path and conversions during result movement reduce round trips through shared memory. N-dimensional DMA performs layout-aware transfers, easing the cost of tensor tiling and rearrangement.

The disclosed per-AI-core storage includes 512 KB L1, 64 KB each for L0A and L0B, 256 KB L0C and 512 KB Unified Buffer. Global accelerator L2 reaches 128 MB. Its 512-byte lines are divided into 128-byte sectors, allowing small irregular accesses without always moving a full line. Software can use allocation hints and residency controls. Coherence across the two dies does not eliminate locality: the white paper explicitly preserves local affinity, so scheduling and tiling should still avoid unnecessary cross-die movement.

Seventy-two 112 Gb/s SerDes lanes form eighteen $\times 4$ ports. This is 56 GB/s raw per port per direction, or 1,008 GB/s across all ports; Huawei quotes 2,016 GB/s bidirectional. Four ports can instead implement PCIe 5.0 $\times 16$, and two ports can implement two 400 Gb/s UBoE interfaces. Those modes consume shared physical resources. A design cannot add maximum UB, PCIe and Ethernet figures as if each came from a separate set of pins. Optics may further constrain off-enclosure speed.

STARS 2.0 schedules heterogeneous engines and synchronizes their work. The CCU executes collective-communication tasks without dedicating the same AI cores to every transfer and reduction. URMA provides asynchronous remote-memory/message operations; UB Memory provides synchronous load/store/atomic semantics. These are complementary mechanisms. The published 128 TB access capability of a chip interface, the 8,192-card architectural domain and a particular SuperPoD's advertised address space are separate limits with separate scopes.

Atlas 350 demonstrates why chip maxima must be translated into board topology. Its 112 GB HBM runs at 1.4 TB/s, and its board power is at most 600 W. A four-card full mesh uses three $\times 4$ UB ports per card and advertises 318 GB/s bidirectional; a two-card arrangement uses four ports for 424 GB/s. Their one-direction equivalents are 159 and 212 GB/s. Both are much smaller than the full chip I/O maximum, because the card exposes a different port allocation and physical link configuration.

10 960 and later: roadmap revisions and optical systems

The 2025 announcement named 950PR for Q1 2026, 950DT for Q4 2026, and a later 960 generation. It described an Atlas 950 design point of 8,192 accelerators across 160 cabinets. The July 2026 WAIC disclosure instead describes a particular 1,024-card Atlas 950 with 1 EFLOPS FP8, 2 EFLOPS FP4 and a 256 TB unified address space. The later white paper still supports an 8,192-card architectural domain. These numbers describe a roadmap design point, a concrete system configuration and an architecture limit; they need not be forced into one product row.

At HUAWEI CONNECT 2026, 960DT readiness moved forward by three quarters to Q1 2027, and 960PR by one quarter to Q3 2027. The company also named 970 and 980 for 2028 and 2029. Atlas 960E couples 4,096 NPUs with Hi-ONE near-packaged optical engines and liquid cooling. Its advertised system peaks are 8 EFLOPS FP8 and 16 EFLOPS FP4 with up to 1 PB of HBM. A system announcement and testing milestone should be tracked separately from silicon readiness and broad customer availability.

11 Scale-up links and memory semantics

Interconnect lineage and normalized directions · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
Kunpeng 920 D2D	2019	n/d	n/d	400 GB/s quoted aggregate; direction n/d
Ascend 910 HCCS	2019	n/d	n/d	240 Gb/s = 30 GB/s quoted link; direction n/d
UnifiedBus 1.0	2025	n/d	n/d	n/d
UnifiedBus 2.0 / 950	2026	112 Gb/s	4	56 raw; 1008 aggregate/device
Atlas 350 UB	2026	n/d	4	53 from card bidirectional rating
950 UBoE	2026	n/d	n/d	50 raw per 400G port
2026 UB optical device	2026	1.6 Tb/s per port	n/d	200 raw/port; direction basis of 280T total unspecified

Interconnect lineage and normalized directions · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
Kunpeng 920 D2D	n/d	Compute / I/O chiplet links	Processor package	Coherent on-package fabric
Ascend 910 HCCS	3	Board/system dependent	n/d	Accelerator communication; do not assume general CPU coherence
UnifiedBus 1.0	7 transceivers per 910C die	Two-tier switched; 7 subplanes	384 NPUs	Peer access; pooled CPU/NPU memory
UnifiedBus 2.0 / 950	18	nD mesh / Clos	8,192 NPUs	URMA; load/store/atomic; on-package coherence separately documented
Atlas 350 UB	3 in 4-card mesh; 4 in 2-card mode	Full mesh / paired	4 / 2 cards	Card UB links
950 UBoE	2	Ethernet fabric	Cluster architecture >128K cards	UB over Ethernet; consumes shared ports
2026 UB optical device	176	Inter-cabinet optical fabric	System-dependent	UnifiedBus; advertised RTT 2 μs

A shared address range is only one part of a memory system. The protocol must also define ordering, atomics, cacheability, synchronization, permissions and failure behavior. Ascend 950 explicitly documents coherent caches across its compute dies and CPU/AI subsystem; that does not establish uniform CPU-style cache coherence across every SuperPoD endpoint. URMA, UB Memory and UBoE should be evaluated by their own semantics and the software paths that use them.

12 DPUs, NICs and the AI communication path

Network products and infrastructure offload · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	GA / milestone	Status	Class	Ports × speed	SerDes
SP681	Current portfolio; GA n/d	Shipping / available	NIC	2 ×25GE	n/d
SP670	Current portfolio; GA n/d	Shipping / available	NIC	2 ×100GE	n/d
SP680 / SP623Q	Current portfolio; GA n/d	Shipping / available	NIC / SmartNIC	4 ×25GE	n/d
SP923Q / SP900	Current product brief; GA n/d	Shipping / available	DPU	4 ×25GE	n/d
QingTian card	2025 documented system	Shipping / available	Cloud infrastructure DPU	n/d	n/d
SP230	WAIC 2026	Announced; GA n/d	Standard NIC	25-200GE family; port count n/d	n/d
SP560 AI NIC	WAIC 2026	Announced; GA n/d	AI NIC	800G advertised; port mapping n/d	n/d

Network products and infrastructure offload · 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Host interface	Programmable engine	Embedded CPU	On-card memory
SP681	PCIe 3.0 ×8	n/d	n/d	n/d
SP670	PCIe 4.0 ×16	Hi1822; engine ISA n/d	n/d	n/d
SP680 / SP623Q	PCIe 4.0 ×16	Hi1822; DPU Solution ecosystem	n/d	n/d
SP923Q / SP900	PCIe 4.0 ×16	DPAK	Hi1822; 24 cores	32 GB DDR4-2400; 4 channels
QingTian card	n/d	n/d	n/d	n/d
SP230	n/d	n/d	n/d	n/d
SP560 AI NIC	UnifiedBus + PCIe	Programmable protocol; ISA n/d	n/d	n/d

Network products and infrastructure offload · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Transport / RDMA	Congestion control	Key offloads	Power
SP681	RoCE v2	PFC / ETS	Checksum; virtualization; network offload	n/d
SP670	RoCE v2	PFC / ETS	Virtualization and network offload	n/d
SP680 / SP623Q	Ethernet; RoCE depends on model	PFC / ETS	OVS / DPI / virtualization ecosystem	n/d
SP923Q / SP900	Ethernet; storage/network offload	PFC / ETS	Bare-metal network/storage; management	n/d
QingTian card	VPC / Ethernet; UB integration by platform	n/d	Virtualization, I/O offload, isolation, trust	n/d
SP230	Ethernet	n/d	Host networking	n/d
SP560 AI NIC	RDMA	n/d	Direct NPU/GPU memory access; collective submission	n/d

Huawei has three distinct networking roles. A conventional or smart NIC connects a host and may offload packet processing. A DPU moves infrastructure services and trust boundaries away from the tenant CPU. An AI NIC prioritizes accelerator communication, direct memory access and collective submission. A network switch and a UB scale-up switch are separate products again. Sharing HiSilicon IP or software interfaces does not make these categories interchangeable.

The SP923Q is a concrete SP900-family DPU: a 24-core Hi1822, four DDR4-2400 channels with 32 GB ECC memory, PCIe 4.0 $\times 16$ and four 25GE ports. Two small M.2 drives support the local operating environment; DPAK exposes network, storage and compute acceleration. The brief specifies an 8 A at 12 V supply requirement, which is not a declared 96 W TDP. Its host link also has more raw capacity than four 25GE ports, illustrating that PCIe provision and external network throughput are different constraints.

The 2026 SP560 announcement adds an 800G AI NIC with UnifiedBus and PCIe host attachment. Direct access to accelerator memory and programmable protocol handling are the disclosed differentiators; the public event record does not provide a full port map, SerDes inventory, embedded-core count or board power. Its 800G label therefore remains a family-level advertised rate, not an invented two-port or four-port configuration. It is also distinct from storage products that happen to reuse the SP560 identifier.

13 Ethernet switching, optics and KV storage

CloudEngine 16800 introduced a high-density 400GE datacenter platform in 2019; the 2023 CloudEngine 16800-X disclosure advanced the portfolio toward 800GE. Current XH16800 documentation describes a chassis fabric, cell switching, ECMP, AI ECN and PFC deadlock prevention. These functions manage congestion and forwarding across an Ethernet fabric. They should not be confused with the number of ports on one switch ASIC or with the bandwidth of a UB accelerator domain. The XH16800 document's "evolution to 800GE" wording is retained rather than treating every listed line card as already 800GE.

The September 2026 UnifiedBus system announcement introduces a 176-port, 1.6T-per-port optical interconnect device and a UBG switch with radix up to 1,024. The first multiplication yields 281.6 Tbit/s of advertised port rate, consistent with the rounded "280T" claim; it does not define bidirectional payload switching capacity. Nor should the second number be read as 1,024 such ports on one monolithic ASIC. Cabinet, chassis, chip and network topology remain separate accounting levels.

Hi-ONE is specified at 7.2 Tbit/s per optical engine with an integrated light source. The 960E comparison replaces 48,000 conventional 800G optical modules with 5,500 engines and claims more than 550 kW of savings. This is Huawei's topology-level comparison, not a measured per-NPU TDP improvement. NPO places optics near the package; it should not automatically be renamed co-packaged optics. The design motivation is to shorten high-speed electrical reach and reduce cabling and power while preserving a large switched domain.

OceanStor M900 adds a PB-scale KV-cache tier, described by Huawei as L3.5 and reachable in one UnifiedBus hop. "L3.5" is a system-tier label, not a CPU cache level. A useful mental model is local HBM for active execution, CPU memory for larger working sets, and a shared persistent or semi-persistent KV tier for reusable context. Storage can avoid recomputation and improve capacity economics, but latency-sensitive decoding still depends on how much useful state reaches HBM on time.

14 System generations and deployment boundaries

From server nodes to SuperPoD and SuperCluster · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
Atlas 2019 systems	2019	Configuration-dependent Ascend nodes	HCCS-connected nodes
Atlas A2	2023-era	Server/module-dependent	HCCS; system-dependent
CloudMatrix384 / Atlas 900 A3	2025	192 CPUs + 384 NPUs; 48 nodes	384 NPUs
Atlas 950 2025 design point	2025 roadmap for 2026	8,192 NPUs; 160 cabinets	8,192 NPUs
Atlas 950 WAIC configuration	2026-07	1,024 NPUs	1,024 NPUs
Atlas 850E	2026-07	Up to 96 cards	96 cards
Atlas 960E	2026 announcement; 960 chips 2027	4,096 NPUs	4,096 NPUs
Kunpeng SuperPoD update	2026-09	Up to 4,096 nodes	256 TB unified pool disclosed

From server nodes to SuperPoD and SuperCluster · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
Atlas 2019 systems	910 integrates 2 ×100Gb/s Ethernet; 25 GB/s raw aggregate	n/d	System-dependent
Atlas A2	n/d	n/d	Air / liquid by product
CloudMatrix384 / Atlas 900 A3	Dedicated RoCE per NPU die; rate n/d	n/d	Liquid-cooled SuperPoD
Atlas 950 2025 design point	UBoE / RoCE; port allocation dependent	n/d	Liquid
Atlas 950 WAIC configuration	n/d	n/d	Liquid
Atlas 850E	n/d	n/d	Air; VCE phase-change technology
Atlas 960E	UB network / RoCE; per-NPU rate n/d	n/d	Full liquid; NPO
Kunpeng SuperPoD update	n/a	n/d	n/d

A maximum supported cluster is not an installed cluster. The 2026 disclosures describe very large scale-out topologies, including hundreds of thousands of NPUs and a multi-rail path to one million. Their engineering significance is the hierarchy and reach of the interconnect; the report does not treat those limits as verified customer installations. Likewise, an addressable memory pool is not necessarily the sum of all physical DRAM in every node.

15 Software evolution and research feedback

Software capabilities aligned with hardware

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Release / layer	Date	Hardware enabled	Key features
CANN / MindSpore launch	2019	Ascend 310 / 910	Graph compilation, operators, runtime and framework
CANN 3.0	2020	Ascend	Broader developer interfaces and operator optimization
Ascend C	Current documentation	Ascend AI cores	Explicit tiling, queue/buffer management, pipelined compute and transfer
CloudMatrix-Infer	2025	CloudMatrix384	Prefill/decode/cache separation; expert parallelism; INT8 kernels
CANN / Mind open source	2025-2026	Ascend	Compiler/runtime/kernel ecosystem opening; framework integration
DaVinci 3 / STARS 2.0 / CCU	2026	Ascend 950	SIMD/SIMT; NDDMA; collective offload; cache hints; low-precision formats
DPAK / QingTian / openEuler	Generation-dependent	Kunpeng / SP900 / cloud	Host OS, network/storage offload and tenant isolation

Ascend C exposes the machine’s memory hierarchy directly enough that kernel authors must understand tiling, buffer lifetime, queues and synchronization. The payoff is precise overlap between data movement and compute; the cost is that porting an operator may involve more than translating its arithmetic. A framework adapter alone does not guarantee equivalent scheduling, collective behavior or quantization quality. CANN, optimized kernels, HCCL-style collectives and inference-serving software have to mature together.

The academic record adds mechanisms rather than a second marketing timeline. ASPLOS 2025’s component roofline explains operator bottlenecks. ISCA 2026’s ENEC optimizes lossless weight compression for Ascend, including vector-friendly decoding and dependency reduction. MICRO 2025’s RICH Prefetcher explores storing richer prefetch information in memory to trade capacity and bandwidth for latency hiding. ENEC is software/research on Ascend; RICH is related CPU research. Neither establishes that a specific new decompressor or prefetcher shipped in Kunpeng or Ascend.

16 Bandwidth hierarchy and derived ratios

Bandwidth by era: scopes are intentionally separate · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Era / reference	On die / die-to-die	Memory
2015-2018 / 310	Explicit local buffers; numeric rate n/d	LPDDR; board-dependent
2019-2022 / 910	L2 fabric 4 TB/s; per-AI-core 128 GB/s read + 128 GB/s write	~1.2 TB/s HBM2
2023-2025 / 910C	Dual compute die; D2D rate n/d	8 memory stacks; R02 rate n/d
2026 / maximum 950	2 AI + 2 I/O; D2D rate n/d	PR 1.6; DT 4 TB/s
2026 / Atlas 350 card	28-Cube bin; D2D rate n/d	1.4 TB/s

Bandwidth by era: scopes are intentionally separate · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Era / reference	Host attachment	Scale-up per accelerator	Scale-out per accelerator
2015-2018 / 310	PCIe; n/d	n/d	n/d
2019-2022 / 910	PCIe 4.0	3 ×240 Gb/s HCCS; direction n/d	2 ×100 Gb/s Ethernet = 25 GB/s raw aggregate
2023-2025 / 910C	Kunpeng / UB	7 UB transceivers per die; rate n/d	Separate RoCE interface per die; rate n/d
2026 / maximum 950	PCIe 5.0 ×16: nominal 64 GB/s/direction	1008 GB/s raw/direction before port reuse	2 ×400G UBoE: 100 GB/s raw aggregate; shared ports
2026 / Atlas 350 card	PCIe 5.0 ×16: nominal 64 GB/s/direction	159 GB/s/direction at 4 cards; 212 at 2 cards	Server NIC-dependent

Derived balance metrics, not measured application performance

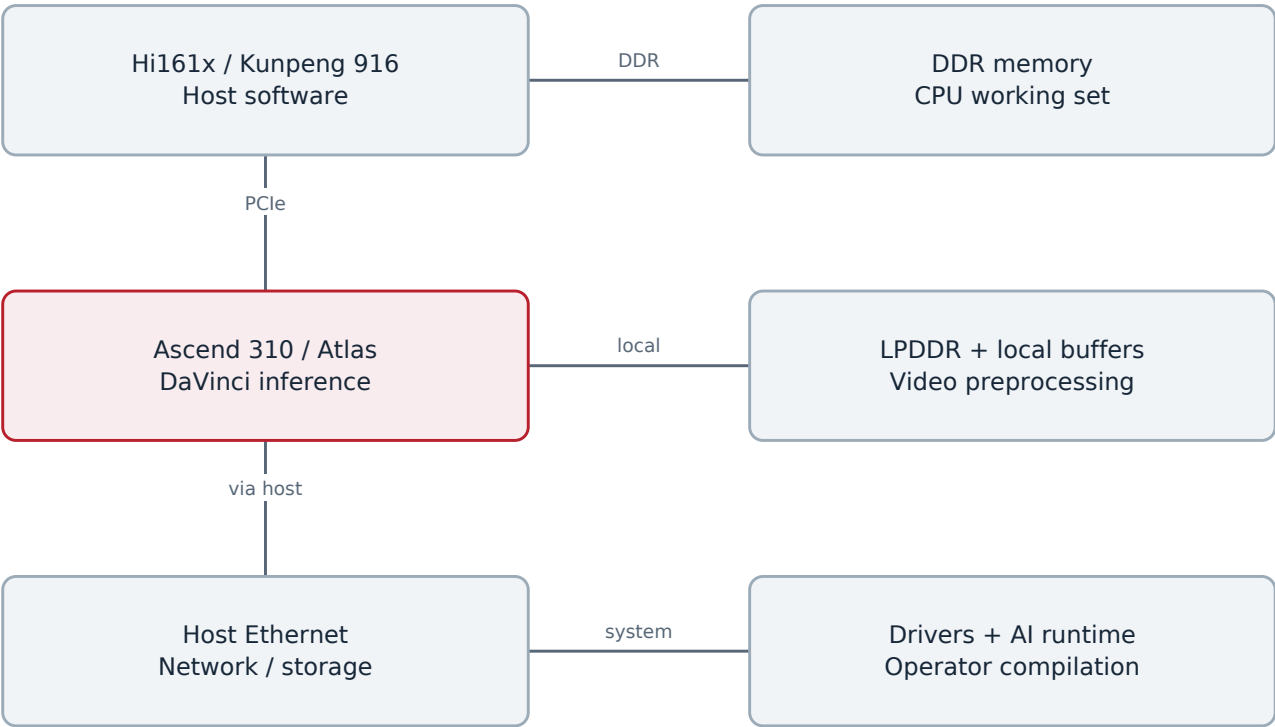
n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Reference	Calculation	Result	Interpretation / boundary
Kunpeng 920	8 × 8 × 2.933 / 64	2.933 GB/s/core	Theoretical DDR4 payload; controller/workload efficiency excluded
Kunpeng 920	64 MB / 64	1 MB/core	Average shared LLC budget, not private allocation
Ascend 910	1.2e12 / 256e12	0.00469 byte/FP16 FLOP	FP16; not a BF16 comparison
950PR max	1.6e12 / 432e12	0.00370 byte/BF16 FLOP	Cube-only dense peak
950PR max	1.6e12 / 865e12	0.00185 byte/FP8 FLOP	Cube-only dense peak
950DT max	4e12 / 486e12	0.00823 byte/BF16 FLOP	2.22× PR bandwidth per BF16 FLOP
950DT max	4e12 / 973e12	0.00411 byte/FP8 FLOP	Cube-only dense peak
Atlas 350, 4 cards	159 / 1400	0.114	One-way card scale-up / HBM; ~11.4%
Atlas 350	378 TFLOPS / 600 W	0.630 TFLOPS/W	BF16 Cube peak / maximum board power; not measured efficiency

17 Integrated stack across four eras

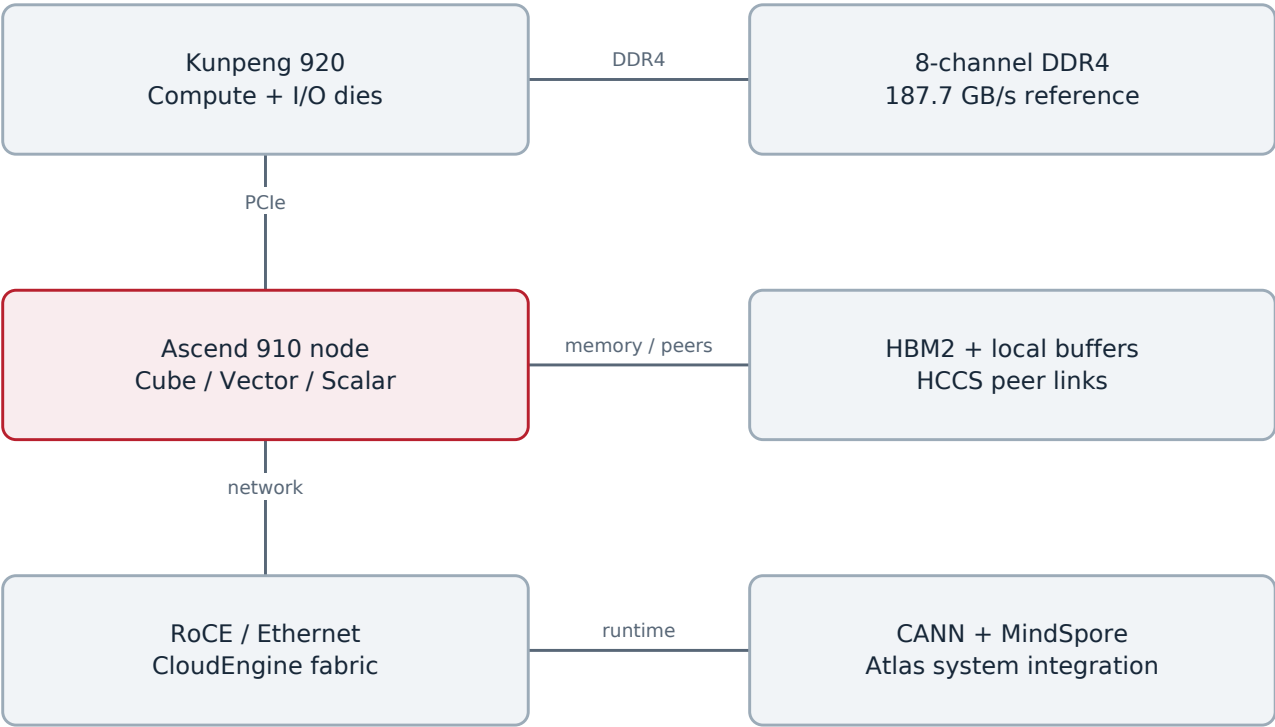
The diagrams show deployment roles and traffic paths. Lines indicate connectivity, not equal bandwidth, universal coherence or simultaneous use of every optional interface. Source coverage follows the preceding family chapters.

2015-2018: host-led inference



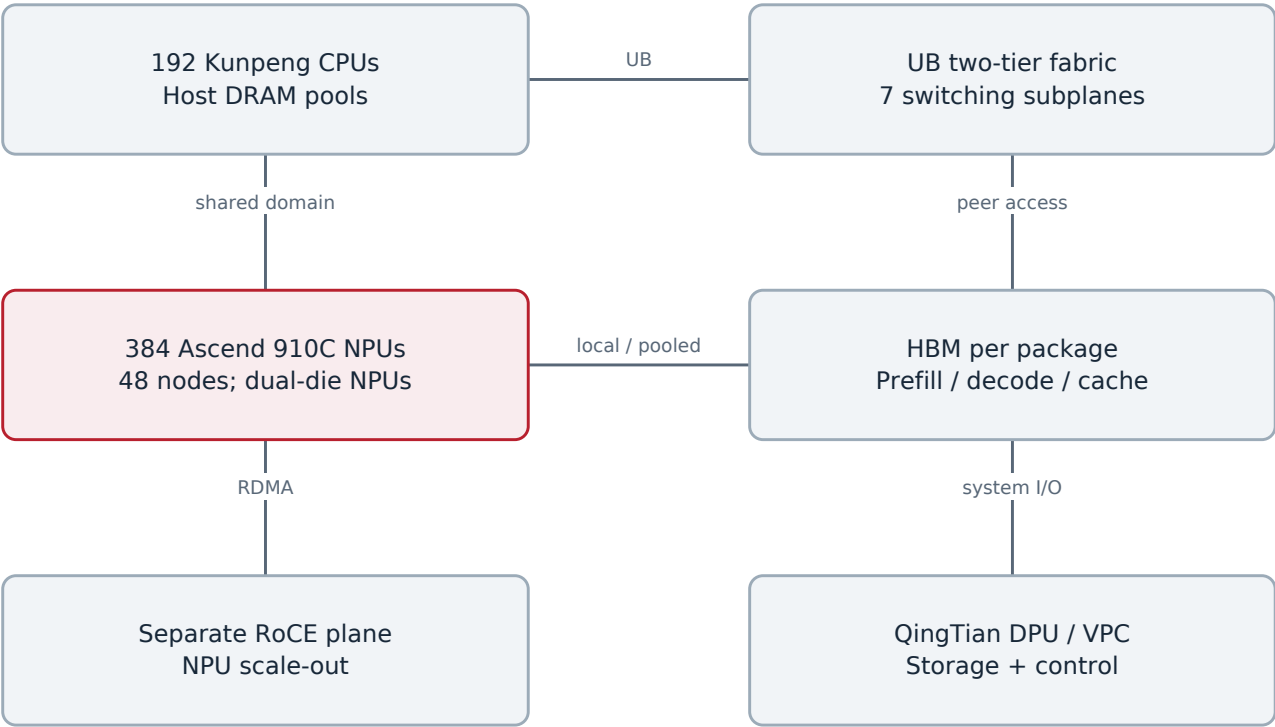
A small NPU augments a server or edge host; this is not yet a large shared accelerator domain.

2019-2022: chiplets and accelerator nodes



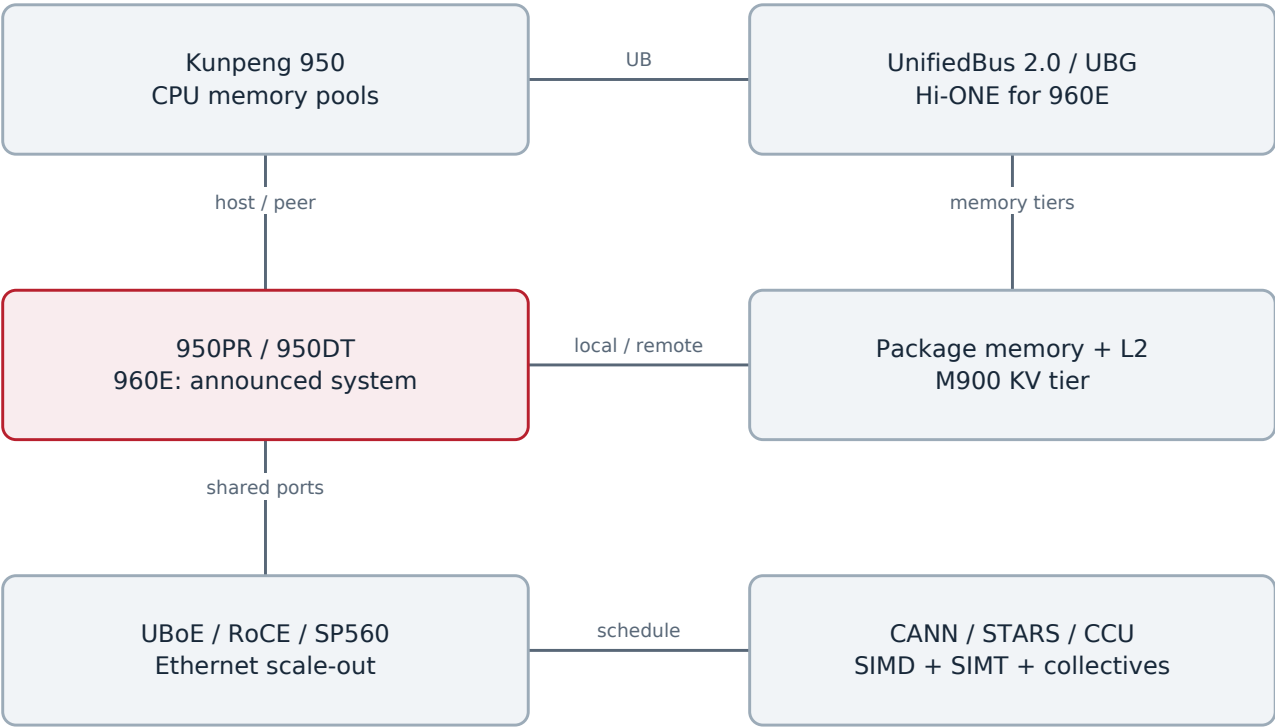
Reusable silicon blocks coexist with explicit accelerator data movement; scale-out remains a network problem.

2023-2025: CloudMatrix384 resource pooling



The UB, RoCE and VPC planes have different roles; CloudMatrix-Infer maps serving stages across the pooled resources.

2026 disclosures: differentiated NPUs and optical scale



This combines product roles, not a claim that every listed component ships in one SKU. 960 chips remain on the 2027 roadmap.

18 Engineering trends and disclosure gaps

Chiplet reuse predates the current AI boom in Huawei's portfolio. Kunpeng 920 used separate compute and I/O dies; the first 910 paired a large compute die with I/O and HBM; 950 makes the compute/I/O partition more explicit while varying its memory subsystem by workload. The benefit is modularity and the ability to optimize different die functions separately. The cost is package wiring, coherent data movement and another layer of placement sensitivity. No public source reviewed here establishes a universal 3D-stacked SRAM design across these shipping products.

Low precision raises compute faster than it raises the supply of useful bytes. The 950 matrix path doubles throughput from BF16 to FP8 and approximately doubles it again to MXFP4. Quantization, vector processing and data layout become increasingly important as the arithmetic gets cheaper. More HBM bandwidth helps memory-bound phases, but operator fusion and keeping intermediate data in L0/L1/Unified Buffer can remove transfers altogether. The next bottleneck may therefore be vector work, synchronization or communication rather than the Cube array.

The PR/DT split is quantitatively meaningful. DT's maximum matrix BF16 throughput is only about 12.5% above PR's, but its memory bandwidth is 2.5 times as large. It therefore offers roughly 2.22 times the memory bandwidth per BF16 FLOP. That is consistent with a training/decoding branch that must sustain more movement per unit of arithmetic. It is an architectural interpretation of the disclosed ratios, not a guarantee that every decode workload runs 2.22 times faster.

Larger scale-up domains give mixture-of-experts models more aggregate memory and more choices for expert placement. They also increase the importance of fabric bisection bandwidth, tail latency, optics reliability and failure containment. A domain of thousands of NPUs cannot be judged by multiplying chip peak alone. The strongest evidence would combine measured model throughput and latency, full system power, recovery behavior and the exact topology under the same workload. These measurements are not uniformly disclosed across Huawei's generations.

The architecture gains more from avoiding unnecessary movement than from renaming all resources "memory." Pooling expands capacity and creates another scheduler's degree of freedom, while also introducing queueing, placement, durability and recovery concerns. Optical integration reduces electrical reach and cable burden, but requires thermal, mechanical and serviceability engineering. These are system-level trade-offs, and should be evaluated with workload traces and failure behavior rather than a single aggregate bandwidth number.

The implementation record is asymmetric. Early Hot Chips and the Kunpeng technical paper reveal die partitioning and core details. ISSCC provides independently identifiable wireline circuit disclosures, including a Huawei 60 Gb/s PAM4 ADC-DSP transceiver, but no reviewed primary link proves that exact circuit is the SerDes in a named Kunpeng or Ascend product. Recent NPU documentation exposes programming and memory mechanisms well, while foundry nodes, yield, package capacity, transistor counts and many power figures remain undisclosed. Manufacturing constraints should therefore be discussed as constraints, not filled with unattributed die-analysis claims.

Opening software and publishing a protocol can broaden adoption, but ecosystem compatibility remains an engineering property. Kernel coverage, compiler quality, collective behavior, device management and the licensing/implementation status of an interconnect all matter. Huawei's combination of CANN, its own system fabric, cloud services and partner boards supplies an increasingly complete stack. Its differentiation is the coordination of these layers; the unresolved comparison is how much useful workload performance and operational resilience that coordination delivers per unit of cost and power.

19 Conference-to-product index · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Year	Paper / talk / disclosure	Presenter / authors
Hot Chips 31	2019	DaVinci: A Scalable Architecture for Neural Network Computing	Heng Liao; Jiajin Tu; Jing Xia; Xiping Zhou
HPCA industry track	2021	Ascend: a scalable and unified architecture for ubiquitous deep neural network computing	Heng Liao et al.
IEEE Micro journal	2021	HiSilicon Kunpeng 920: First 7-nm Chiplet-Based 64-Core Server CPU With Arm Ecosystem	Jing Xia; Chuanning Cheng; Xiping Zhou; Yuxing Hu; Peter Chun
ISSCC	2019	6.2: 60Gb/s PAM-4 ADC-DSP Transceiver; 7nm; adaptive power scaling	M.-A. LaCroix et al., Huawei Ottawa
ASPLOS	2025	Squeezing Operator Performance Potential for the Ascend Architecture	Yuhang Zhou et al.
MICRO	2025	RICH Prefetcher: Storing Rich Information in Memory to Trade Capacity and Bandwidth for Latency Hiding	Ningzhi Ai et al.
ISCA	2026	ENEC: A Lossless AI Model Compression Method Enabling Fast Inference on Ascend NPUs	Jinwu Yang et al.
R-CCS Symposium	2020	Kunpeng platform presentation	Zhaohui
Huawei Developer Conference	2025	CloudMatrix384 cloud service	Huawei Cloud
HUAWEI CONNECT	2025	SuperPoD, 950/960 roadmap, UnifiedBus 2.0, CANN opening	Eric Xu; Yang Chaobin; Zhang Dixuan
MWC Barcelona	2026	SuperPoD and AI infrastructure showcase	Huawei
WAIC	2026	Atlas 950, Atlas 850E, SP560 AI NIC, compute modules	Huawei
IEEE ISCAS	2026	Tau scaling and logic folding	Huawei researchers
HUAWEI CONNECT	2026	960E, Hi-ONE, Peerium, M900, Kunpeng pools	David Wang; Huawei

Verified disclosures and related research - 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Product	Evidence type	Source
Hot Chips 31	Ascend 310 / 910	Architecture, die/package, memory, links	C01
HPCA industry track	Ascend / DaVinci	Product architecture; vendor reprint read	C04 / P01
IEEE Micro journal	Kunpeng 920	Core, cache, die partitioning; journal, not MICRO conference	P01 / C02
ISSCC	Wireline IP; product adoption unproven	Circuit; program reports 6.9 pJ/b at 32 dB loss	C05
ASPLOS	Ascend	Operator characterization and optimization research	R01
MICRO	Related CPU research	Research; no confirmed Kunpeng adoption	R06
ISCA	Ascend 910B2 study	Model compression research	R03 / R05
R-CCS Symposium	Hi161x / Kunpeng 916 / 920	CPU roadmap and platform; historical future targets	C03
Huawei Developer Conference	CloudMatrix384	Cloud/system availability	E17 / R02
HUAWEI CONNECT	Kunpeng / Ascend / UB	Architecture roadmap and ecosystem	E02 / E09 / E16
MWC Barcelona	Atlas / Kunpeng	System and industry deployment	E05
WAIC	Ascend / networking	System and product launch	E18 / E13
IEEE ISCAS	Related scaling research	Research direction; not ISSCC or a product node disclosure	E06
HUAWEI CONNECT	AI infrastructure stack	System launch, roadmap update, optics/storage architecture	E01 / E03 / E04 / E10 / E11 / E12

The index is a verified conference-to-product map, not a claim to enumerate every Huawei-affiliated publication. No reviewed primary disclosure establishes a generation-by-generation Kunpeng/Ascend product paper at DAC, VLSI, IEDM, ECTC, OFC, SC or ISC. Likewise, Computex, OCP, GTC, re:Invent, Ignite and Google Cloud Next are not automatically Huawei product disclosure channels. They remain search targets rather than invented entries. MWC, WAIC, HUAWEI CONNECT and Huawei's developer events have direct, useful records for this portfolio.

20 Names, reuse and items that must not be merged

Product and architecture crosswalk

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Name / identifier	Meaning	Important boundary
Hi1620 / Kunpeng 920 / TaiShan V110	Original 2019 CPU / product / core	New-model 920 platforms have separate specifications
Kunpeng 950 / Linx816	Host CPU family / embedded Ascend CPU core	Do not equate the two microarchitectures
DaVinci / Ascend / Atlas	Architecture / processor / hardware system	One architecture spans multiple chips and boards
910C / Atlas 900 A3 / CloudMatrix384	NPU / SuperPoD product / cloud-system architecture	The paper abbreviates the processor to Ascend 910; launch identifies 910C
950PR / Atlas 350	Prefill/recommendation processor / 28-Cube PCIe card	128 GB chip maximum vs 112 GB card configuration
UB buffer / UnifiedBus	AI-core local memory / system interconnect	Same acronym, different layers
HCCS / HCCL	Hardware interconnect / collective software library	A library name does not specify link bandwidth
SP600 / SP900 / SP560 AI NIC	NIC family / DPU family / AI NIC	Do not infer the AI NIC from a similarly named SSD
Hi-ONE / NPO	Optical engine / near-packaged optics	Not a blanket synonym for CPO
OceanStor M900 / L3.5	KV storage system / system-tier label	Not a CPU L3 cache or on-chip SRAM

Source map

- 01 Scope and executive synthesis - C01, E01, E02, E03, E11, P01, P08, P14, P15, R02
- 02 Reading the numbers and product names - C01, C04, P01, P14, P15, R01, R02, R03, R05, R06
- 03 Host CPU generations - C03, E02, P01, P08, P09, P10, P13, P16, R07
- 04 2015-2019: from server Arm to Kunpeng 920 - C03, P01, P10, P16, R04, R07
- 05 Updated 920 and Kunpeng 950: platform evolution - E02, E11, E12, P08, P09, R02
- 06 Ascend generation comparison - C01, E02, E03, E07, E11, E18, E20, P01, P12, P14, P15, R01, R02, R05
- 07 2018-2021: DaVinci and the first Ascend systems - C01, C04, E07, E20, P01, P12
- 08 A2 to A3: accelerator nodes become a compute pool - E02, P12, R01, R02, R05
- 09 Ascend 950: computation, memory and communication co-design - P14, P15
- 10 960 and later: roadmap revisions and optical systems - E02, E03, E11, E12, E18, P15
- 11 Scale-up links and memory semantics - C01, E02, E04, P01, P14, P15, R02
- 12 DPUs, NICs and the AI communication path - E13, P04, P05, P06, P07, P15, R02
- 13 Ethernet switching, optics and KV storage - E03, E04, E10, E11, E14, E15, P11
- 14 System generations and deployment boundaries - C01, E02, E03, E05, E07, E11, E12, E18, P12, R02
- 15 Software evolution and research feedback - E08, E09, E18, E20, P01, P03, P04, P07, P08, P15, R01, R02, R03, R05, R06
- 16 Bandwidth hierarchy and derived ratios - C01, P01, P14, P15, R02
- 17 Integrated stack across four eras - C01, E03, E11, E13, P01, P15, R02
- 18 Engineering trends and disclosure gaps - C01, C05, E03, E09, E11, E12, E17, P01, P08, P15, R01, R02
- 19 Conference-to-product index - C01, C02, C03, C04, C05, E01, E02, E03, E04, E05, E06, E09, E10, E11, E12, E13, E16, E17, E18, P01, R01, R02, R03, R05, R06
- 20 Names, reuse and items that must not be merged - C01, E02, E03, E11, E13, P01, P04, P06, P08, P09, P12, P14, P15, R02

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